

Lesson 7: Capacitors

Charge, voltage, time, and stored energy

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

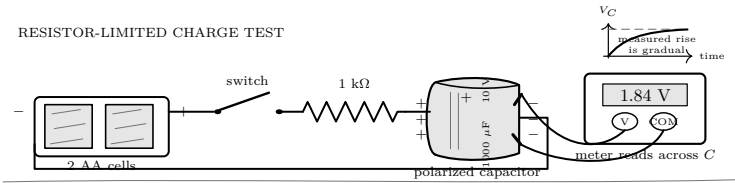
What you need

- 2 AA cells
- 1 kΩ resistor
- 1000 μF capacitor rated 10 V or higher

- Multimeter
- Switch

Predict first

Will capacitor voltage jump instantly to the source value through a resistor?



Investigate

1. Adult confirms capacitance, voltage rating, and polarity. Begin discharged through the resistor.
2. Connect battery–resistor–capacitor. Measure voltage across the capacitor every five seconds.
3. Disconnect the battery and discharge only through the resistor while recording voltage.
4. Repeat with a second resistance; never short the leads with metal.

Explain from the evidence

Capacitance is $C = Q/V$. The resistor limits current, so voltage changes over time; larger R or C produces a longer time scale. Ideal stored energy is $E = \frac{1}{2}CV^2$, showing why voltage matters quadratically. This course uses only small low-voltage components, yet treats even them as charged until measured and resistor-discharged.

Change one thing

Plot voltage against time for two resistors on the same axes. Identify what changed and what did not.

Evidence record			
Prediction	_____	One change	_____
Observation	_____	Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral [electricity research library](#).